ISPRS Journal of Photogrammetry and Remote Sensing 228 (2025) 370–387

Contents lists available at ScienceDirect

ISPRS Journal of Photogrammetry and Remote Sensing

journal homepage: www.elsevier.com/locate/isprsjprs

UC–Change: a classification-based time series change detection technique for improved forest disturbance mapping using multi-sensor imagery

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ARTICLE INFO

Keywords:
Forest monitoring
Time-series change detection
Forest fire mapping
Salvage harvest mapping
Sentinel–2
Landsat MSS

ABSTRACT

Unsupervised Classification to Change (UC–Change) is a versatile technique that detects forest disturbances in satellite images by analyzing changes in the spatial distribution of spectral classes over time. This approach can fully utilize the spectral resolution of individual sensors without requiring atmospheric correction or radiometric normalization. Resulting multisensor capabilities set UC–Change apart from established time-series change detection methods, such as Continuous Change Detection and Classification (CCDC), LandTrendr, Composite2Change (C2C), and Global Forest Change (GFC). With the growing number of Earth observation satellites, the ability to utilize diverse datasets is increasingly important for extracting information relevant to sustainable natural resource management. The algorithm's effectiveness is demonstrated using a dataset containing 275 Landsat and Sentinel–2 images acquired over a forested area in British Columbia, Canada, from 1972 to 2020. The 100 km × 100 km study site has been actively harvested in recent decades and experienced many wildfires and a mountain pine beetle (MPB) outbreak. The spatio-temporal accuracy of clearcut and fire-scar detection was assessed using the Vegetation Resources Inventory (VRI) and National Burned Area Composite (NBAC) products, respectively, and compared against the C2C 1985 – 2020, CCDC 2002 – 2019, and GFC 2001 – 2022 maps available online. Overall, the UC–Change algorithm detected 85.2 % of the reference VRI 1974 – 2018 cutblock pixels at a temporal resolution of ± 1 year (90.3 % at ± 3 years). It detected 86.0 % of 1985 – 2018 VRI pixels, outperforming C2C (58.8 %). For the period 2002 – 2018, UC–Change mapped 87.1 % of the reference cutblock pixels, exceeding C2C (54.5 %), CCDC (74.0 %), and GFC (70.4 %). UC–Change, C2C, CCDC, and GFC detected 71.0 %, 54.6 %, 37.4 %, and 67.2 % of 2006 – 2018 reference forest fire pixels, respectively. UC–Change provided improved forest harvest and fire-scar detection in areas heavily affected by the MPB outbreak and forests characterized by low canopy cover. It represents a new, fundamentally different approach to time-series analysis, suitable for independent use or in concert with existing methods.

1. Introduction

The goal of forest disturbance mapping is to determine how, where, and when changes in forest cover occurred, as well as provide information on the temporal dynamics of such changes. Only time-series change detection techniques, which use large, temporally continuous (typically one or more images per year) sets of satellite data, can provide this information over large areas. After the opening of the Landsat archive in 2008, automated time-series algorithms have largely replaced traditional bi-temporal (i.e., comparing image pairs) and multitemporal (i.e., comparing several images) techniques as time-series analysis provides a basis for a deeper understanding of forest changes and trends (Hansen and Loveland, 2012; Banskota et al., 2014, Wulder et al., 2018; Wulder et al., 2022).

Accurate mapping of disturbances over large areas requires satellite imagery with a resolution under 100 m (Mantovani and Setzer, 1996;

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https://doi.org/10.1016/j.isprsjprs.2025.07.028
Received 5 August 2024; Received in revised form 5 July 2025; Accepted 20 July 2025
Available online 24 July 2025
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Pape and Franklin, 2008; Potapov et al., 2011; De Sy et al., 2012; Banskota et al., 2014). For many geographical regions (Zhang et al., 2022), Landsat satellites have provided such data since 1972, with sufficient spatial and temporal resolution to identify small clearcuts (10 – 20 ha) and other forest disturbances. The freely available Landsat archive and improvements in image quality have made time-series analyses more feasible. Recent imagery, like Sentinel-2, further enhances data quality, necessitating advanced multi-sensor methods to extract valuable information.

There are currently three main approaches to time-series change detection using Earth observation data: classification-based, trajectory-based, and hybrid. Classification-based routines either compare a series of classification maps (Knudby et al., 2010) or classify entire stacks of multitemporal images together (Schroeder et al., 2011). For example, Schroeder et al. (2011) used 16 growing-season Landsat 5 TM and Landsat 7 ETM + images from 1987 to 2008. They manually delineated training areas representing clearcuts and fire scars for every image in the time series, as well as persistent forest, persistent non-forest, and water classes. Techniques like these can be very flexible as they can use auxiliary information and data acquired during any part of the growing season, any number of spectral bands, sensors, and data types without atmospheric correction or data normalization.[1] However, they typically require a considerable amount of user input, are subjective and lacking in consistency across images, and are not practical for larger time series or areas.

Change detection based on temporal trajectory analysis is a more popular approach because it is more automated and can be used to not only detect forest disturbances, but also monitor post-disturbance recovery (Kennedy et al., 2010; Zhu and Woodcock, 2014; Hermosilla et al., 2015a). This approach analyzes how pixel values change over time. Trajectory-based techniques extract information about forest disturbances by fitting simple models in the form of straight lines or polynomial curves to segments of the temporal profiles of individual pixels representing pre-disturbance, disturbance, and post-disturbance states.

LandTrendr (Kennedy et al., 2010) is one of the most established trajectory-based algorithms (Kennedy et al., 2018). For every year in the time series, the algorithm creates a single mosaic from a number of midsummer images using only clear-sky pixels. After creating such image composites, LandTrendr identifies straight-line segments in a time series of normalized pixel values. The algorithm then removes vertices deemed unimportant or caused by noise. Based on the slope of individual segments, the algorithm identifies not only abrupt events (i.e., forest disturbances), but also long-term trends (e.g., forest regrowth). An example of such segmentation is presented in Fig. 1A, where LandTrendr identified segments in a time series of Normalized Burned Ratio (NBR) values. NBR is one of the most widely adopted spectral indices for mapping forest disturbances (Huang et al., 2010; Hermosilla et al., 2015b; Hughes et al, 2017).

Composite2Change (C2C) is another technique based on temporal trajectory analysis (Hermosilla et al., 2015b; White et al., 2017). It follows the same approach as LandTrendr with a few minor differences. For example, LandTrendr starts with a single segment and gradually increases the number until a maximum number of segments, defined by the user, is reached, or a good enough fit between the segments and data values is achieved (the threshold for which is also user-defined). In contrast, C2C starts with the maximum number of segments and iteratively reduces their number. Both algorithms have change-attribution extensions that classify detected disturbances into harvest, fire, and other classes (Hermosilla et al., 2015a; Kennedy et al., 2015).

The Continuous Change Detection and Classification (CCDC) algorithm also detects changes by analyzing the temporal trajectories of individual pixels; however, unlike LandTrendr and C2C, CCDC uses both summer and winter observations (Zhu and Woodcock, 2014). By fitting harmonic models to all available observations, CCDC can predict normal seasonal variations in pixel values. Any lasting deviations from the modelled trends are labeled as forest disturbances (Fig. 1B).

All trajectory-based techniques, including LandTrendr, C2C, and CCDC, share a number of limitations. They are sensitive to spectral band dissimilarities among sensors, bidirectional reflectance distribution function (BRDF) effects, the availability of cloud-free and snow-free observations, and interannual variations in plant vigor and phenology (e.g., the length of the growing season). While the impact of the former two factors can be reduced by using normalized data, such as Harmonized Landsat and Sentinel-2 (HLS) data[2] (Claverie et al., 2018; Shang et al., 2022; Zhu et al., 2022), the latter two factors present a more difficult challenge (Arevalo et al., 2022a). Although curve-fitting algorithms (e.g., CCDC) can utilize much more data than line-fitting algorithms (e.g., LandTrendr and C2C) because they are not limited to the peak of the growing season, they also *need* more data (seven cloud- and snow-free observations per year, according to Zhang et al., 2021) to generate accurate time-series models. Therefore, clouds can present a problem for this type of technique too, in addition to snow cover in the winter (Zhu and Woodcock, 2014). Considering that the number of usable observations available for the period 1972 – 1998 is very low for many parts of the world, this is a serious limitation (Zhang et al., 2022; Landsat Archive Dashboard: landsat.usgs.gov/landsat-archive-dashboard). Consequently, existing global forest change maps do not cover that period (Friedl et al., 2022; Hansen et al., 2013; Zhang et al., 2022). A more flexible approach is required to produce forest change maps for geographical regions and periods of time with reduced data availability.

The Global Forest Change (GFC) map, produced using a hybrid approach, shows forest gain and loss for every year between 2000 and present (Hansen et al., 2013). It is the only forest change map that has a 30-m spatial resolution and is updated yearly for the entire planet (GFC, 2023). Various modifications of the algorithm have been developed (Hansen et al., 2008; Potapov et al., 2011, 2012, 2015). According to Potapov et al. (2015) the core part of the GFC technique is classification-based change detection, but instead of classifying a stack of images, a diverse set of multitemporal metrics derived from a Landsat time series is used for classification (e.g., the slope of linear regression between reflectance value and observation date). As the technique also analyzes the temporal trajectories of individual pixels, it is considered to be hybrid. This approach has the advantages of both post-classification and pixel-based methods; however, it also has some of the disadvantages, such as requiring training data and atmospheric correction and/or radiometric normalization of data (Potapov et al., 2015; Potapov et al., 2020).

This paper presents a new classification-based technique named Unsupervised Classification to Change (UC–Change), which was developed to address some of the limitations of the aforementioned techniques and introduce new capabilities. UC–Change combines the flexibility of classification-based routines with the high automation and ability to track forest recovery usually associated with trajectory-based techniques. It can use multi-sensor data without compromising the unique spectral information provided by each individual sensor. UC–Change can use any number of images per year acquired during any part of the growing season. This flexibility should enable more effective utilization of satellite data in areas and time periods with reduced data availability. In addition, unlike trajectory-based methods (Zhu et al.,

[1] Atmospheric correction may improve results when there is spatial variation of atmospheric conditions (e.g., haze) within individual images. However, these variations are typically minor and do not have a substantial effect on image classification, as was the case in this study. Temporal variation of atmospheric conditions should not affect classification-based change detection techniques because these techniques do not compare pixel values directly.

[2] It is noteworthy that the HLS dataset currently does not include data from Landsat 7 and earlier Landsat missions (see further discussion in §2.1).

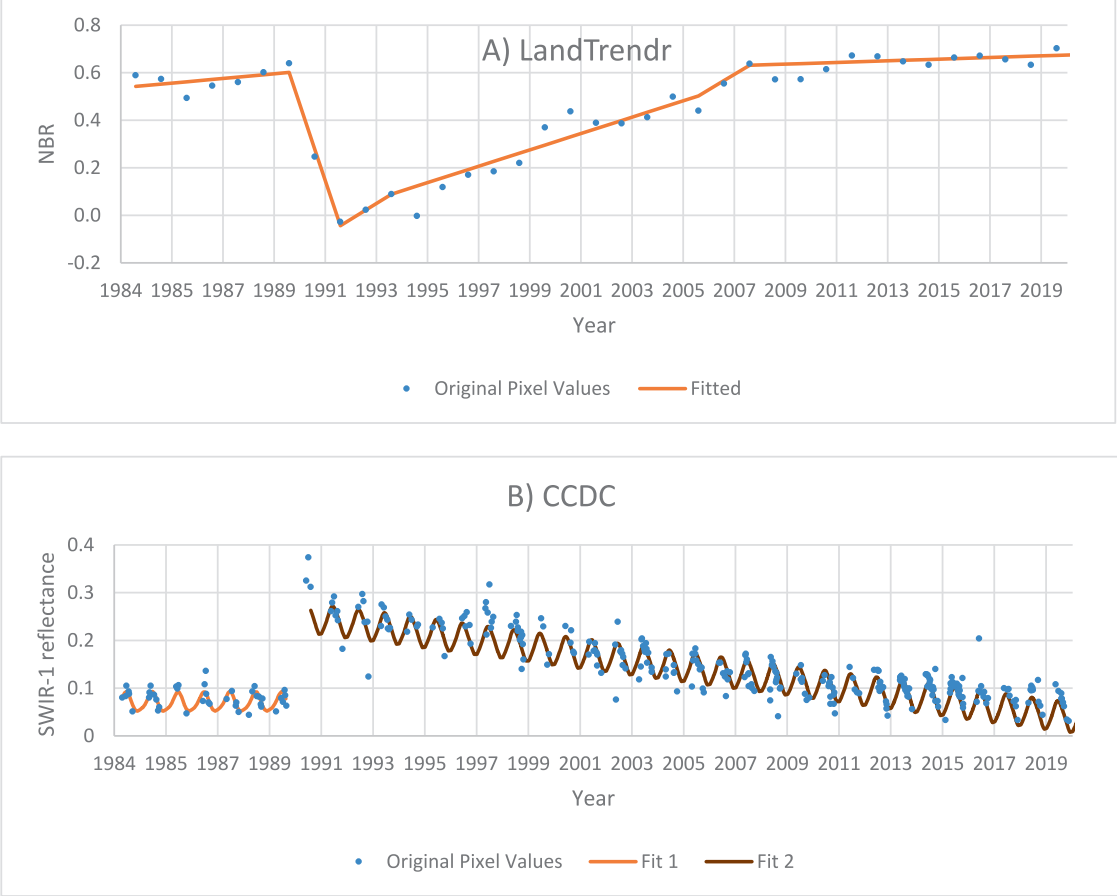


Fig. 1. Temporal segmentation results for a pixel in a 1991 clearcut in British Columbia, Canada (pixel location: 53°13′53″N, 124°13′28″W), produced using the default LandTrendr and CCDC parameters (Kennedy et al., 2018; Arévalo et al., 2020). A) Mid-summer NBR values (blue) and LandTrendr segmentation results (orange). B) SWIR-1 surface reflectance values for all cloud- and snow-free Landsat observations (blue) and CCDC harmonic models representing the pre- and post-disturbance states (orange and brown, respectively). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2020), UC–Change does not require atmospheric correction or radiometric normalization because it does not compare pixel values directly. This is particularly advantageous for incorporating sensors with different spectral characteristics or those lacking readily available atmospherically corrected data, such as the Landsat 1 – 5 MSS sensors (Chen et al., 2020). Consequently, depending on the availability of MSS data, UC–Change can map forest disturbances dating back to 1972, preceding the acquisition of more commonly used Landsat TM data by at least a decade. These capabilities collectively enhance the accuracy, efficiency, and temporal extent of forest change monitoring, enabling more informed decision-making for ecosystem management and conservation.

The study compares UC–Change, C2C, CCDC, and GFC maps for a complex mosaic of forest disturbance in western Canada. The primary goal is to introduce UC–Change and discuss its technical aspects. The comparison highlights its strengths and weaknesses relative to the other change detection techniques, with comprehensive validation confirming its effectiveness. Future research directions are proposed given the success of this new approach.

2. Study area and data

A study area located in British Columbia, Canada, was selected to introduce the new technique based on several criteria, such as the availability of independent reference data for the entire 1972 – 2020 period, sufficient forest disturbances to test the temporal and spatial accuracy of the produced maps, and presence of different forest types in terms of species composition and stand density. The area corresponds to the grid square T10UDD in the Military Grid Reference System (MGRS[3]) and is 100 km by 100 km in size with the center at 52°45′N, 123°40′W (Fig. 2). It is located south of the city of Prince George and lies within the Montane Cordillera ecozone (Wiken 1986; Leckie et al., 2016). The dominant tree species is lodgepole pine (*Pinus contorta* var. latifolia), followed by white spruce (*Picea glauca*) and subalpine fir (*Abies lasiocarpa*). The southern part, corresponding to the Sub-Boreal Pine – Spruce (Very Dry Cold) subzone in Fig. 2, has a low canopy cover of approximately 30 % (VRI, 2023), which is still sufficient for land to be classified as forest based on the definition of forest adopted by Natural Resources Canada (NRCan, 2023a). The terrain is characterized as gently rolling (Leckie et al., 2016). A mountain pine beetle (MPB) outbreak in the 2000 s killed most lodge-pole pine trees in the area (BC Ministry of Forests, 2023a). Major forest fires occurred in 2006, 2009, 2010, 2014, 2017, and 2018, with an especially large fire in 2017.

2.1. Input data

The main dataset comprised 200 Landsat 5 TM, Landsat 7 ETM+, Landsat 8 OLI, and Sentinel–2 images acquired between 1984–06-22 and 2020–09-10 (Fig. 3). Landsat Collection 1 Level-1 (terrain corrected

[3] UC-Change uses an MGRS-based tiling system identical to the one used for Sentinel-2, to avoid Sentinel-2 data resampling from one UTM zone to another and re-mosaicking.

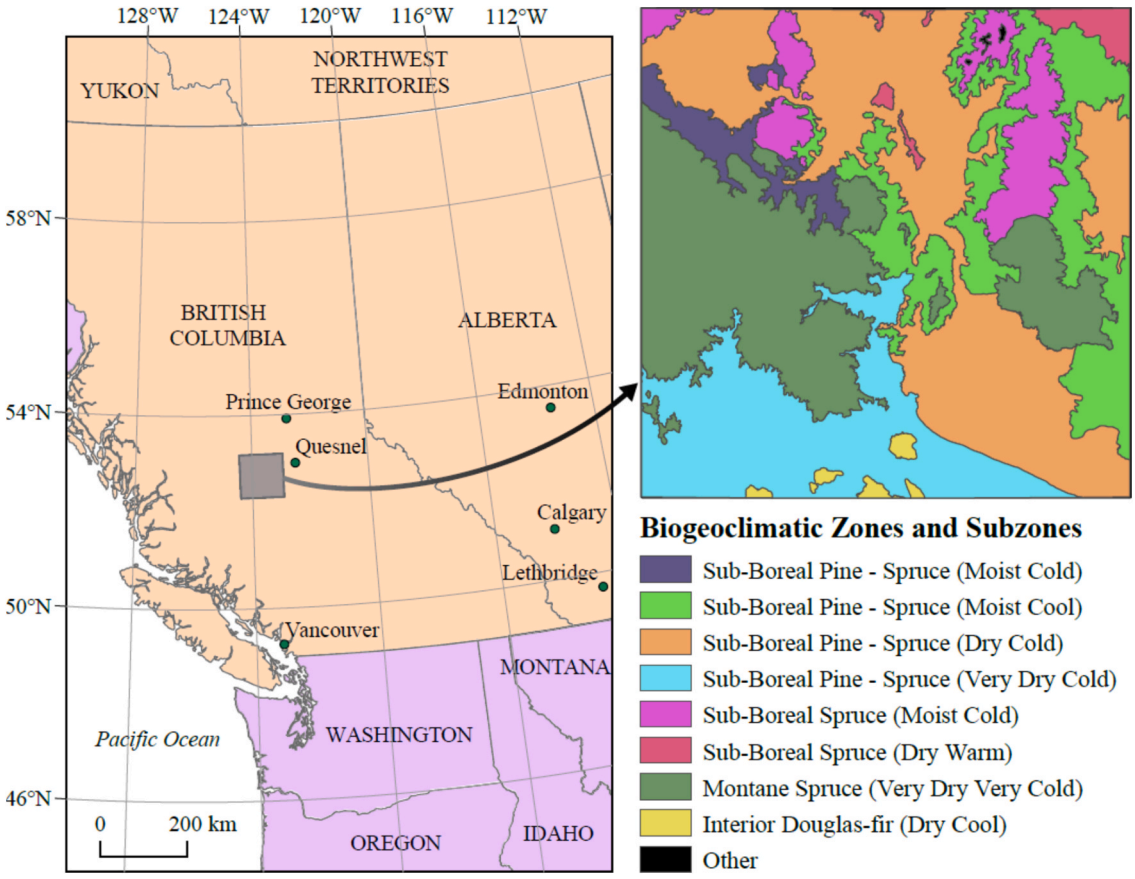


Fig. 2. Left: location of the 100 km × 100 km study area in British Columbia, Canada. Right: biogeoclimatic zones and subzones (in brackets) in the study area (based on BEC, 2020).

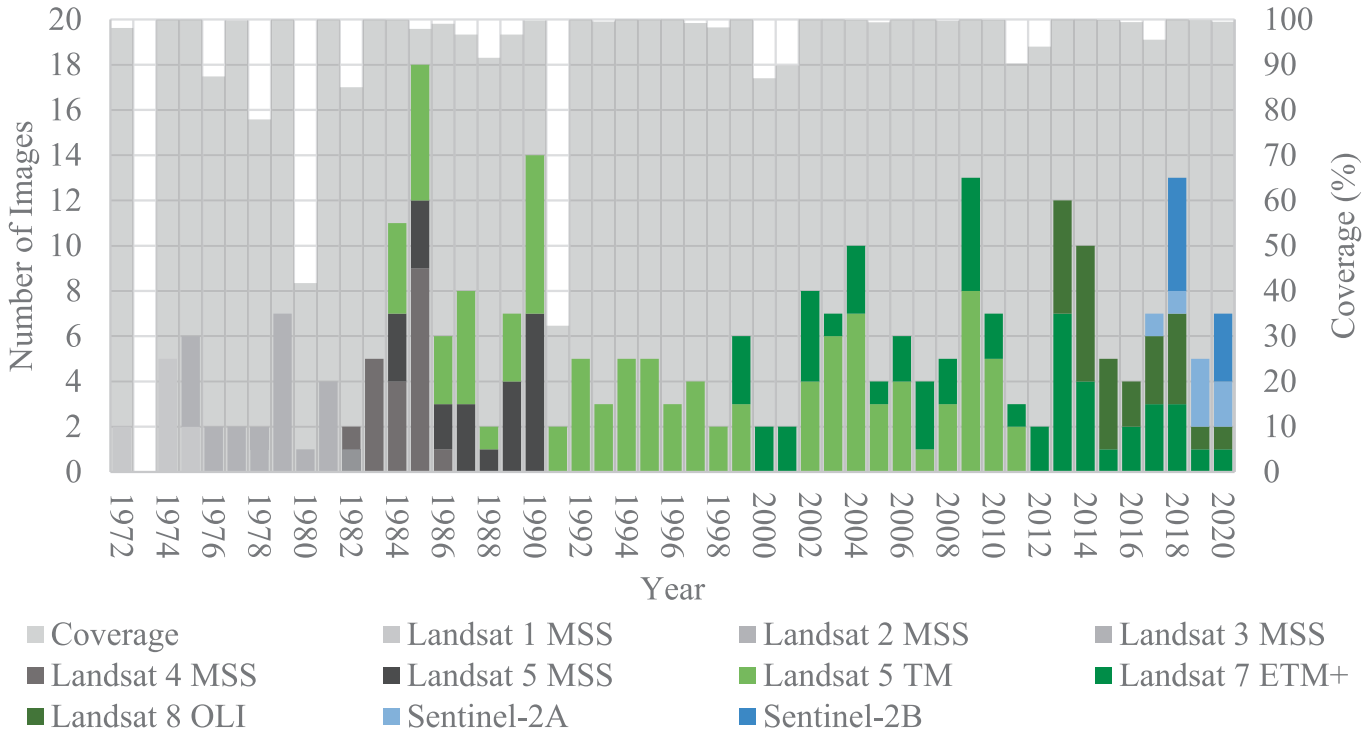


Fig. 3. Number of images per year per sensor used in the study. The light grey bars in the background represent study area coverage (percent area covered per year) provided by the data. Note: some images were not available or were discarded due to high geometric errors, resulting in the data gaps in the chart 1973, 1980, 1991.

but not atmospherically corrected) and Sentinel–2 Level-1C (*ortho*-rectified, top-of-atmosphere reflectance) data were selected based on cloud cover. Preliminary tests using atmospherically corrected data yielded change maps of approximately the same accuracy. To reduce the potential impact of unmasked cloudy pixels and the processing time, only images with $\leq$ 50 % cloud cover were selected, except for particularly cloudy years, identified as when the 50 % threshold resulted in $<$ 90 % yearly coverage of the study area. For those years, the threshold was reduced to 25 % cloud cover to maintain a high level of yearly coverage.

The "Quality" band included with Landsat and Sentinel–2 images was used to mask clouds. Two additional steps were required for the Sentinel–2 data due to the lower quality of the cloud masks available during the time of the study. Firstly, a blue-band threshold (top-of-atmosphere spectral reflectance $>$ 16.5 %) was used to mask most remaining clouds. The threshold was determined empirically to ensure that no clearcuts, which also have a high blue reflectance, were masked together with clouds. Secondly, a mask for cloud shadows was produced by shifting the cloud mask according to the mean solar azimuth angle at the time of image acquisition and an offset of 1,800 m. The value (1,800 m = distance of 60 Landsat pixels) accounts for low-level clouds ($<$2000 m above ground) and most sun elevation angles (from 38° in September to 59° in June for this study site) possible at the acquisition time of Sentinel–2 data (~11:30 a.m.) in the study area. While this value is not sufficient for mid- and high-level clouds ($>$2000 m), a higher value would result in the removal of a large number of useful pixels. A buffer of 300 m was applied to both Landsat and Sentinel–2 cloud and cloud shadow masks to remove most residual cloudy pixels. Despite such an aggressive cloud masking, using almost four months of data (June 1 – September 20, a period roughly corresponding to the snow-free season) per year still made it possible to achieve a nearly 100 % coverage of the study area for most years. This approach to cloud masking was used as a first solution for rapid and automatic preprocessing of Sentinel-2 data over large areas. High-quality cloud masks have since became readily available for Sentinel-2 data (Pasquarella et al., 2023).

Like many other forest change detection algorithms, UC–Change requires that agricultural areas be masked in input data. Preliminary tests using other areas in Canada showed that unmasked crop fields resulted in many false detections. Due to crop rotation, pixel values can change considerably across multiple years, which may result in crop fields being incorrectly identified as a forest change class by change detection algorithms (Huang et al., 2010). That is why in other studies, croplands were either masked in the input data (Stueve et al., 2011; Hermosilla et al., 2015a), removed after change detection (Zhao et al., 2018), or omitted from accuracy assessment (Hughes et al., 2017).

UC–Change uses existing crop maps to mask agricultural lands and water bodies. While this step is important in regions with extensive croplands, agricultural areas were not a significant source of error in this study. The Agriculture and Agri-Food Canada (AAFC) federal government crop inventory maps of the study area (available for 2011 – 2023) did not classify any areas as specific crop types, and only up to 0.3 % of the land area was classified as "Agriculture (undifferentiated)," depending on the year. Given this minimal coverage, we used a single, high-quality agricultural mask – the AAFC 2017 map (AAFC, 2023) – which had the highest classification accuracy in British Columbia (matched only by the 2018 map). Although using year-specific crop maps could marginally improve change detection accuracy, we chose the 2017 product for its overall reliability. Its higher accuracy was likely due to the availability of Sentinel-2 data, which was used alongside Landsat-8, Gaofen-1, and RADARSAT-2 to generate the inventory (AAFC, 2024).

UC–Change was designed for mapping large areas, which requires careful consideration of processing time and storage, as well as band selection. The blue band contributed little to the detection of stand-replacing disturbances (0.2 % for cutblocks and 3.6 % for fire scars) but increased false detection (from 0.8 % to 1.3 %) in preliminary tests; therefore, this spectral band, as well as bands with lower spatial resolution (e.g., the thermal bands of the Landsat 5 – 8 sensors), were not used to produce the final map. This decision resulted in a selection of five Landsat and nine Sentinel 2 bands (Fig. 4). Using 8-bit raw DN data instead of 16-bit surface reflectance data further reduced storage requirements for Landsat 5 and 7. All four spectral bands of the Landsat MSS sensors were utilized to detect changes.

Because the version of UC–Change used in this study could not yet process data with varying spatial resolutions, the 10–m (visible and near-infrared) and 20–m (red edge and short-wave infrared) spectral bands of Sentinel–2 data were resampled to 30 m using the nearest neighbor resampling method to match the spatial resolution and pixel grid alignment of Landsat data. While alternative approaches such as pixel aggregation could be explored for potential improvements, current efforts are focused on making UC-Change fully agnostic to spatial resolution differences, eliminating the need for spatial resampling altogether.

It is noteworthy that errors of co-registration of Landsat 8 OLI and Sentinel–2 data can exceed 16 m (Stumpf et al., 2018) and, therefore, can introduce new issues, such as incorrect mapping of boundaries between patches of disturbed and undisturbed forest. The HLS project partially addresses this issue by co-registering Landsat and Sentinel-2 data and reprojecting them onto a common 30-m pixel grid (Claverie et al., 2018); however, the HLS image collection does not include Landsat 7 ETM + and prior Landsat sensors. Because the pixel grids of Landsat Collection 2 and HLS data do not align, using a combination of HLS and Collection 2 data would require the spatial resampling of older Landsat data (a shift of half a pixel), which would affect the spectral integrity of the data. Therefore, while the HLS dataset is a valuable resource, some of its components are not yet suitable for direct use in longer time series analyses (e.g., 1972 – 2020, as in this study). In contrast to the HLS project, which resamples both Landsat Collection 2 (using bicubic convolution) and Sentinel-2 data (using area-weighted averaging), we preserved the original Landsat data and resampled only the Sentinel-2 imagery.

In addition to the main dataset, a separate one containing 75 Landsat 1 – 5 MSS images was used to expand the time series to 1972. Only Tier 2 data were available for the study area. Tier 2 data are geometrically and radiometrically corrected but do not meet the Tier 1 quality criterion that requires a Root-Mean-Square Error (RMSE) of less than 12 m geometric correction error. The spatial resolution of Tier 2 data is 60 m, resampled from the original MSS resolution of 68 m $\times$ 83 m by the data provider. Data values represent scaled top-of-atmosphere radiance and reflectance (the scaling factors were not applied in the study because UC–Change does not require radiometric correction of input data). The initial set of 92 images was selected automatically using the same approach that was used for the other sensors; however, based on visual examination, 17 images had to be discarded due to poor cloud masks or large geometric errors. The images were cloud masked the same way as Sentinel–2 images but using a different threshold for clouds (the lowest wavelength available was used, which was the green band, with a top-of-atmosphere reflectance threshold of $>$ 13 %). Produced masks were of lower quality than those generated for the Sentinel–2 data because the green band does not allow a clear separation of clouds and fresh clearcuts. All four MSS spectral bands (green, red, NIR-1, and NIR-2) were used for change detection. Because the Landsat 5 MSS and Landsat 5 TM sensors, which were mounted on the same satellite, provide a temporal overlap between the MSS dataset and the main dataset, the MSS dataset was processed separately to avoid spatial resampling.

2.2. Forest change maps used for comparison

UC–Change maps were compared with maps produced by three methods: C2C, CCDC, and GFC. They represent the three main approaches to time-series change detection that are currently in use, as described in the Introduction section. The CCDC and GFC maps used in

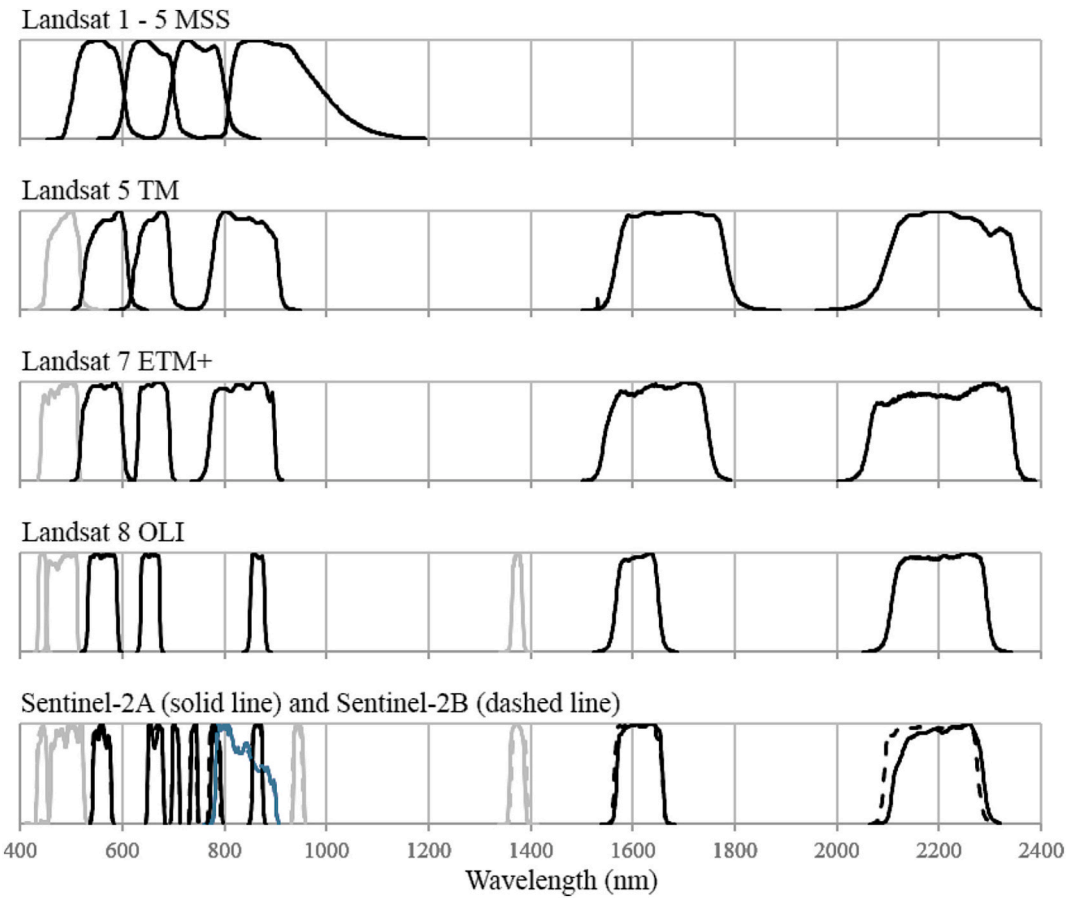


Fig. 4. The spectral response functions of the ten sensors used in this study. The bands that were not used are grayed out. The thermal and panchromatic bands of the Landsat sensors were not used and are not shown. The 10-m NIR band of the Sentinel-2 sensors is shown in blue to visually separate it from the spectrally overlapping 20-m bands. The NIR-2 band of Landsat MSS 4–5 was slightly different from that of MSS 1–3, which is not shown on the graph. Data obtained from the USGS and ESA websites. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

this study were subsets of global maps covering the years 2001 – 2019 and 2000 – 2022, respectively (Arevalo et al., 2022b; GFC, 2023). The available C2C maps encompass the country of Canada and span the period 1984 – 2020 (C2C, 2023). Similar to other maps, the C2C map will be referred to as "C2C 1985 – 2020" because changes that occurred between summer 1984 and summer 1985 are labelled "1985" on this map.

The available C2C and GFC maps are two-dimensional, showing all changes in a single layer instead of multiple layers (i.e., one per year). Therefore, these maps cannot properly represent areas disturbed more than once during the mapped period. The C2C map shows only disturbances with the greatest magnitude of change (C2C, 2023). Based on the analysis of areas affected by multiple disturbances, the GFC map shows the disturbances that occurred first. Although GFC is updated annually, newly released versions (e.g., Version 1.11, which covers 2000 – 2023) add change pixels only for the most recent year (2023) and do not revise or reprocess earlier years (2000 – 2022). As a result, change pixels from earlier versions remain unchanged, and repeat disturbances are not captured. In contrast, the C2C 1985 – 2015 and 1985 – 2020 maps were fully reprocessed and, when converted into yearly maps and combined, allowed at least some repeat disturbances (e.g., pre- and post-2015) to be identified. To account for the limitations associated with using two-dimensional (as opposed to yearly) maps and to make the comparison of the maps fair, all repeat disturbances (i.e., areas disturbed more than once) were removed from the 1985 – 2015 reference data (5.4 % of VRI cutblock pixels and 21.6 % of NBAC forest fire pixels), and 1985 – 2020 repeat disturbances were removed from the 2016 – 2020 reference data (2.9 % of VRI and 12.2 % of NBAC pixels).

2.3. Reference data

Clearcut detection was evaluated using Vegetation Resources Inventory (VRI, 2002; VRI, 2023) forest harvest polygons and the National Burned Area Composite (NBAC, 2023) database was used to assess the detection of wildfire burn scars. A set of polygons representing protected areas was used to estimate false detections (ECCC, 2023). It was assumed that only wildfires and biotic factors, but not harvesting, could cause disturbances in protected areas after they were established.

VRI is the forest inventory data of British Columbia and it contains detailed information on several decades of forest harvesting activities in the province. The VRI cutblock polygons are manually delineated using aerial imagery collected approximately every ten years (Bourgeois et al., 2018). Consequently, these polygons are very precise and accurate spatially, but not always temporally. The temporal information (i.e., the date of harvest) comes from supplementary sources, including data reported by the forestry industry, but the precise harvest date is missing for many cutblocks (i.e., only the year but not the month and day of disturbance are reported). Therefore, some temporal discrepancy between the change detection maps and the VRI dataset was expected. The fact that change detection techniques timestamp change pixels in a way that often does not correspond to the calendar year also contributes to this temporal discrepancy. Because change detection algorithms generally use only mid-summer, growing-season, or snow-free data, they are unable to determine the exact year of change for disturbances occurring in the winter. For example, all changes that occurred in the winter of 2009/2010 would be labeled "2010″ as a result.

To assess the quality of the VRI dataset, 3,545 cutblocks were

manually delineated through consecutive bi-temporal image interpretations of 1984 – 2016 Landsat data. It was determined that the VRI polygons represented approximately 86.3 % of all clearcuts in the study area. 93.7 % of the VRI polygons had a temporal accuracy of ± 1 year. Therefore, the actual accuracy of the change detection maps used in the study is likely higher than the accuracy estimates produced using the VRI data.

There were two reasons why the VRI dataset was chosen over our own set of polygons for the assessment of cutblock detection. First of all, VRI maps were produced by an independent group of analysts using an independent set of data (stereo aerial photographs[4]), and these maps are freely available to the public; thus, they allow an unbiased, reproducible, and transparent evaluation of Landsat-based change detection products. Secondly, VRI polygons show the full spatial extent of individual cutblocks, making it possible to assess the detection of cutblock edges. Finally, the VRI dataset also covers the period from 1972 to 1983, for which it would be challenging to produce reference data of comparable quality using the available Landsat images due to the coarser spatial resolution of the MSS sensors.

National Burned Area Composite (NBAC, 2023) polygons were used to assess the detection of forest fire scars. NBAC polygons are derived from multiple sources, including C2C maps, which were used for comparison in this study. However, verified C2C forest fire objects represented a very small fraction of the NBAC dataset (0.8 % of the area of all NBAC polygons used in this study) – the fraction for which higher quality sources were not available (Skakun et al., 2022). Because C2C-derived NBAC polygons contributed only 0.8 percentage points to the overall C2C fire scar detection results (§4.2), NBAC was deemed suitable for the assessment of all change detection maps, including the C2C map.

Confidence intervals for the NBAC dataset were estimated at ± 5.1 % for the period 2004 – 2020 and ± 13.6 % for the period 1986 – 2003 (Skakun et al., 2022). Hall et al. (2020) reported that fire scar polygons generated from Landsat imagery achieved an average accuracy of 96 % when validated against polygons derived from aerial photos. Only NBAC polygons produced using aerial photographs, and satellite data with a spatial resolution < 55 m, were selected for this study.

We note that NBAC polygons do not indicate variation in burn severity. For example, approximately 10 % (by area) of island remnants (surviving patches of trees) are not represented in NBAC polygons (Hall et al., 2020). Using generalized vector data for the accuracy assessment of highly heterogeneous classes, such as fire scars, in classification maps is known to result in considerably underestimated accuracies (Wulder et al., 2006). For this reason, a relatively lower agreement between the NBAC map and change detection maps was expected. Nonetheless, the NBAC polygons can still be used as the basis for comparison to assess the relative performance of different change detection techniques.

Polygons representing provincial parks and other protected areas were used to estimate false detection of forest harvest activities in the UC–Change and C2C maps (ECCC, 2023). The study area was heavily affected by a mountain pine beetle (MPB) outbreak, including the protected areas (NRCan, 2023b). However, MPB-affected areas were not detected by the UC–Change algorithm due to the lack of abrupt and localized changes in infested areas. Therefore, any change pixels detected within protected areas except those located inside NBAC forest fire polygons were considered false positives in the UC–Change map (Fig. 5). The same rule was applied to the C2C map because it also depicts only harvested and burned areas. The GFC product was also optimized to capture abrupt forest cover changes (Potapov et al., 2019,2022), such as clearcuts and forest fires, which means it may not be as sensitive to the more gradual canopy changes associated with MPB outbreaks. However, no studies have definitively confirmed or refuted the ability of the GFC algorithm to detect MPB damage. Therefore, the false detection results for the GFC map in this study should be interpreted with caution.

Unfortunately, protected areas cannot be used to assess false detection in the CCDC map because some change pixels in that map may actually represent MPB-affected areas. Zhu et al. (2020), who used an algorithm similar to CCDC, reported a 60 % detection rate for the "stress" disturbance category, which included insect damage. Accounting for MPB detection by CCDC in this study would be challenging because available provincial data on MPB, such as the Pest Infestation Polygons dataset (BC Ministry of Forests, 2025), lack sufficient spatial detail.

False detection of cutblocks was evaluated only for the period 2002 – 2015. Lack of protected areas before 1995, and very large forest fires after 2016 made it difficult to produce reliable estimates of commission errors outside this period.

3. UC–Change algorithm

UC–Change detects groups of pixels that become separate from larger groups of pixels in an image compared to previous images. These larger groups of pixels will be referred to as temporally-constrained stable (TCS) pixels. Each image in a time series can have multiple groups (classes) of TCS pixels, and each such group is a collection of pixels that share a similar change vector in a predefined time window (e.g., four years – the temporal constraint). For example, one TCS class may represent a type of stable forest cover, while another may represent cutblocks of similar age.

Areas affected by forest disturbances occurring within the processing time window do not contain TCS pixels because the change vector of these areas deviates from that of their parent group. Therefore, change pixels can be identified by removing pixels that were stable in previous images from the TCS pixels detected in the current image. This process allows for the precise identification of forest changes over time without relying on bitemporal image comparisons or analyzing the temporal trajectories of individual pixels.

UC–Change detects TCS and change pixels through a number of steps involving image classification and reclassification (Fig. 6). Although the algorithm utilizes traditional unsupervised classification and concepts of supervised classification routines, it does not require any user input once the parameters are selected. The following paragraphs describe each processing step in detail.

3.1. Unsupervised classification

To detect TCS and change pixels, original numerical data must be converted into categorical data (i.e., classification maps). The UC–Change algorithm uses maps derived by unsupervised image classification. The choice of classifier should be based on several criteria: 1) disturbed areas and pre-disturbance forest must be classified into different spectral classes; 2) the spatial distribution of spectral classes must remain similar in a sequence of images except for newly disturbed areas; and 3) the classifier should be able to process hundreds of images in an automated and timely fashion. Meeting these criteria, the K–means classifier (Steinhaus, 1956) was found to be well-suited for this study. The Iterative Self-Organizing Data Analysis (ISODATA) algorithm was the only other unsupervised classifier available in the programming environment (ENVI 5.5 + IDL 8.7), and it produced identical results when configured to generate a fixed number of classes.

K–means reclusters pixels through an iterative process until it finds groups that are most separable from one another in the spectral domain. In the first iteration, pixels are classified based on the minimum Euclidean distance to evenly-distributed arbitrary class means. During

[4] The VRI dataset can be confused with the Consolidated Cutblocks dataset, which is also maintained by the Ministry of Forests, Lands and Natural Resources Operations of British Columbia. The latter contains additional cutblock polygons that are missing in the former. These missing polygons are generated using Landsat data (BC Ministry of Forests, 2023b), whereas VRI cutblock polygons are manually delineated using aerial photos (VRI, 2002).

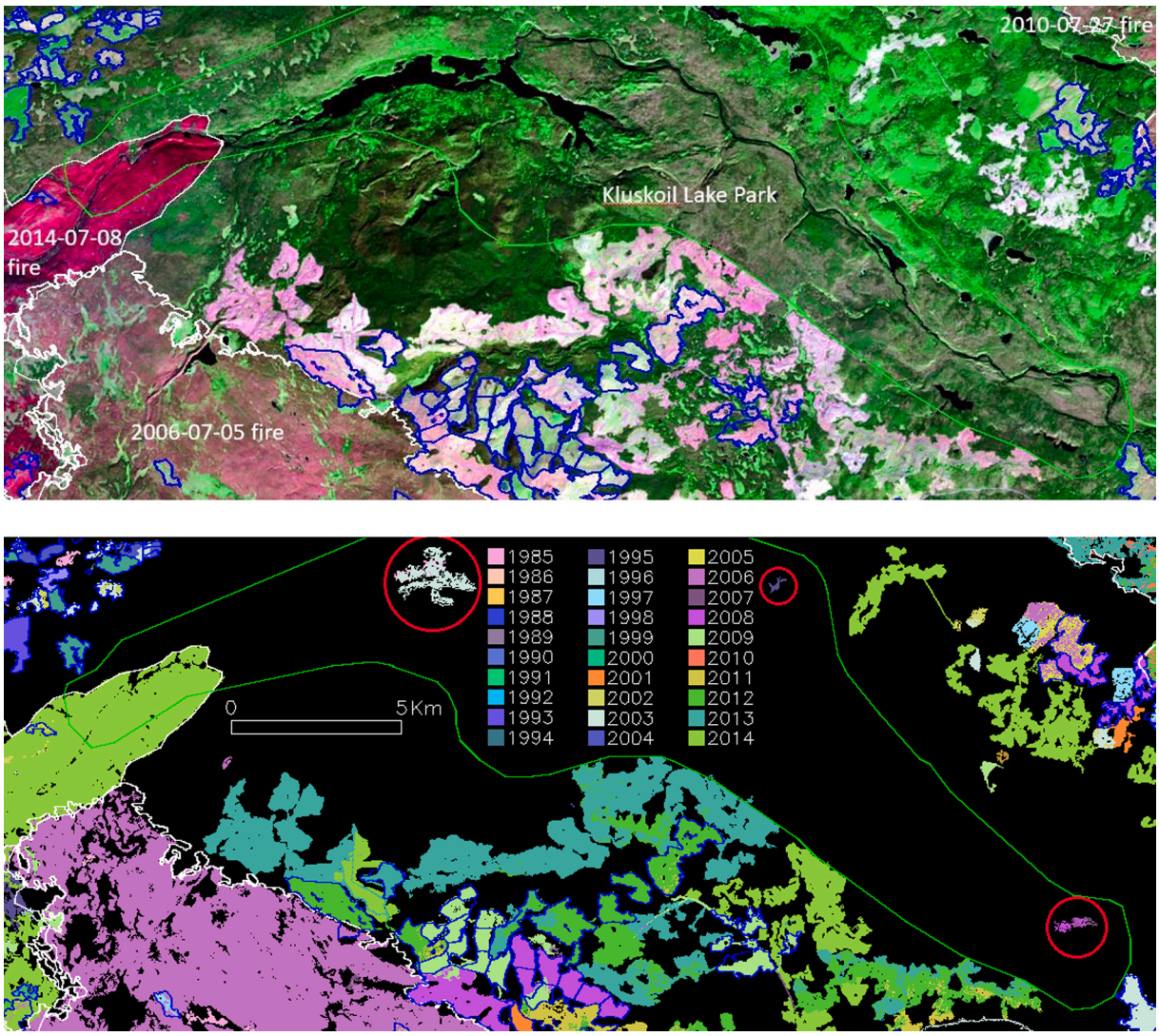


Fig. 5. VRI (blue), NBAC (white), and Kluskoil Lake Provincial Park (green) polygons overlaid on a Landsat 8 image (RGB: red, NIR, SWIR–2) acquired on 2014–09-13 (top) and the UC–Change map (bottom). Colors in the UC–Change map represent the year of disturbance. A large number of false positives (highlighted with red circles) can be seen inside the park, mainly due to changing wetland conditions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

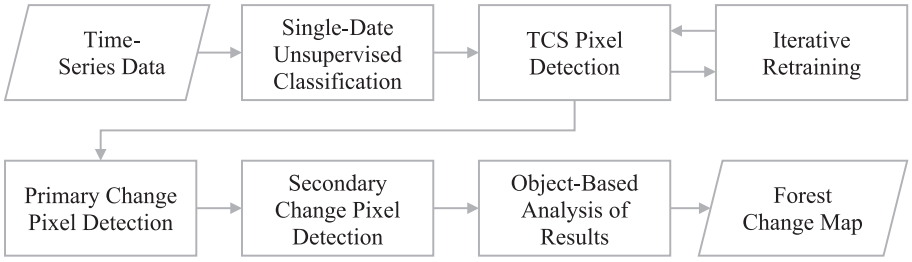


Fig. 6. Flowchart of the UC–Change algorithm.

each following iteration, class means are recalculated and pixels are reclassified based on the same minimum-distance-to-mean rule.

3.2. *Stable pixel detection*

Although the spectral characteristics of land-cover types change through seasons and sometimes years,[5] the spatial distribution of spectral classes stays mostly unchanged through time for any particular

[5] Some land-cover types, such as regenerating cutblocks, grasslands, and wetlands, may exhibit inter-year variability in surface reflectance.

major land-cover type, at least when the number of classes is small. This is true because plants of the same species usually change their spectral characteristics in unison based on the leaf phenological cycle (Lechowicz, 1984; Hill et al., 2010). This is also the case for clearcuts and fire scars. For example, when images were classified into 10 classes, nearly all clearcut pixels remained in one class for at least four years. To extract stable pixels from these 10-class classification maps, the algorithm would locate pixels that continue sharing one class for up to four years (Fig. 7a-c). Therefore, the number of TCS classes corresponds to the number of spectral classes in the k-means maps they are derived from. It is important to note that in order to detect TCS pixels, the UC–Change algorithm matches only the spatial distribution of spectral classes while ignoring cluster labels/numbers.

3.2.1. TCS pixel detection: parameters

TCS pixel detection (and, consequently, the quality of change detection) depends on three inter-dependent parameters: the number of spectral classes, the length of time with the processing window, and the number of images used in TCS detection. The K–means classifier was set to produce 10 classes in this study. The choice of 10 classes was found optimal after a series of preliminary tests on several datasets representing various forest types across Canada. The algorithm could not find enough TCS pixels when using 15-class K–means images, because the distribution of spectral classes varied too much from image to image. In contrast, when it was set to produce five or fewer classes, the K–means classifier could not separate clearcuts and fire scars from some types of undisturbed forest. The most optimal results, with very little variation (less than ± 1 % deviation in producer accuracies), were achieved when images were classified into 6 to 14 classes. In these instances, clearcuts and fire scars were consistently classified into their own distinct class and remained so for the duration of the processing time window, which is the main requirement for the algorithm to work. The middle value (10 classes) was found to work well in other parts of British Columbia and in several test sites in other provinces of Canada.

The maximum length of time of the processing window was set to four years based on two factors: the length of time patches of disturbed forest remained homogeneous in K–means classification maps (i.e., contained only one spectral class) and the availability of spatially overlapping images. The former determines the maximum value for this parameter, while the latter determines the minimum value. In 10-class K–means maps, cutblocks and fire scars (or parts of fire scars) contained only one spectral class for at least four years and often as long as seven years (Fig. 8). Therefore, in order for the UC–Change algorithm to detect as many TCS pixels as possible for disturbed areas, it was determined that in this study area and also in other parts of the province of British Columbia, the length of the processing period should not exceed four years. This and other parameters can be determined, potentially automatically, by examining changes in the distribution of spectral classes within disturbed areas (§3.5, Fig. 8).

UC–Change is an image-based (as opposed to pixel-based) technique and, therefore, when processing a temporal subset of images, it can only utilize the parts of images that overlap. Instead of using all four years of data within the time window, which can contain tens of images, the algorithm selects the *four images* where cloud-free areas have the largest spatial overlap. Using fewer than four images per set increased the probability of detecting unmasked cloud pixels as TCS pixels, while using a larger number of images makes it more difficult for the algorithm to find overlapping images within the processing time window. For the same reason, the time window should not be too small, especially for the 1984 – 1998 period, where only Landsat 5 data were available.

3.2.2. Iterative retraining option

To detect changes accurately, the algorithm must locate as many TCS pixels as possible. While it is not possible to detect all of them due to image noise, unmasked clouds and cloud shadows, and other factors, the number of detected TCS pixels can be increased through a process called iterative retraining, whereby TCS pixels derived from the K–means classification output for a temporal subset of images are used as training

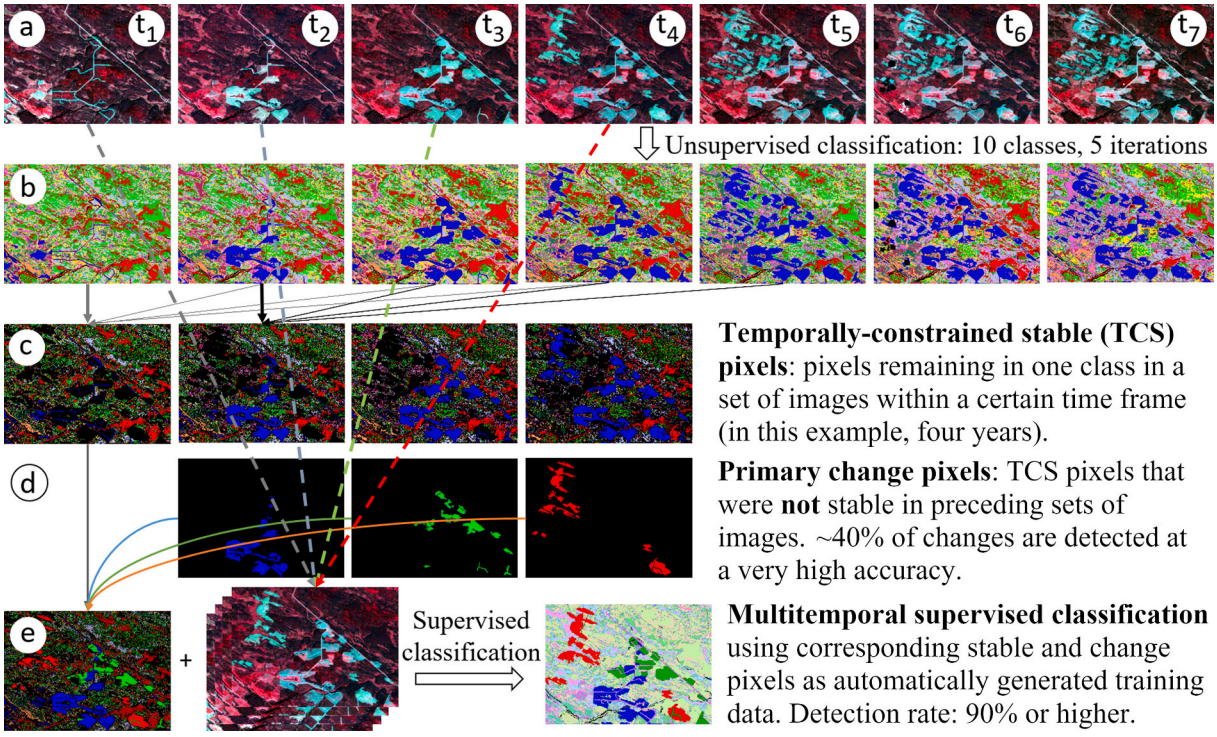


Fig. 7. Main steps of the UC–Change change detection: a) original multispectral images in false color (green–red-NIR; t_1 – t_7 are the acquisition dates of the images; bright areas are fresh clearcuts); b) K–means classification output (10 classes); c) TCS pixels derived from the K–means maps; d) primary change pixels derived from TCS pixels; and e) multitemporal classification using the TCS and change pixels as training samples and a stack of the four corresponding original images together with the final output product. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

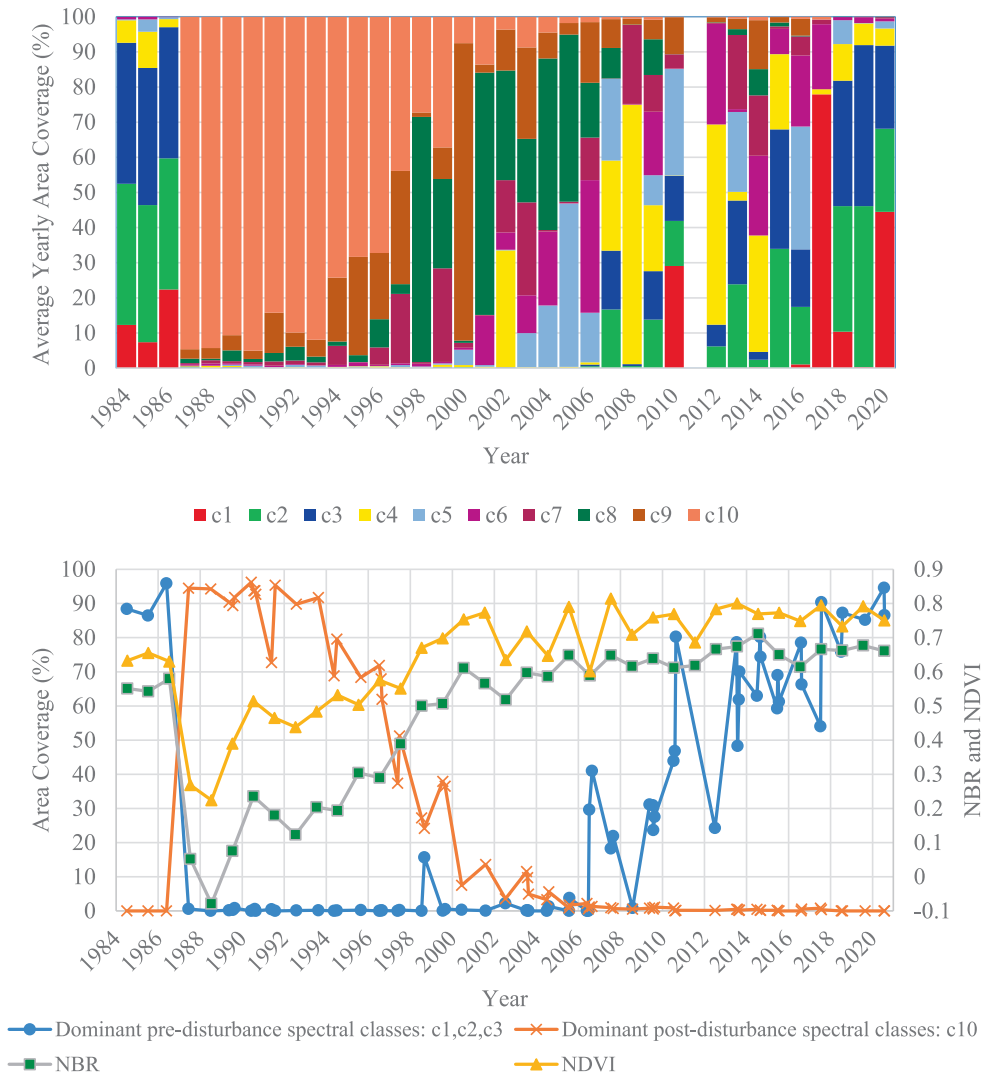


Fig. 8. Top: Changes in the distribution of SWIR-ranked spectral classes in a 1987–07-01 clearcut over time. Bottom: changes in the cumulative percentage of dominant (containing 90% or more pixels) pre-disturbance and post-disturbance spectral classes in the same cutblock. The NBR and Normalized Difference Vegetation Index (NDVI) temporal trajectories of the cutblock are added for comparison.

data in single-date Maximum Likelihood classifications of the images in the subset. For example, TCS pixels detected for each K–means cluster in image X are used as training samples for the Maximum Likelihood classification of the same image, producing a new 10-class map that replaces the original K–means map. The resulting classified images can then be used again to locate TCS pixels, this time in much larger quantities. This process can be repeated multiple times to produce highly consistent maps. The goal is to minimize multitemporal differences in the distribution of spectral classes caused by atmospheric conditions, phenology, and differences in sensor characteristics, and to maximize those caused by forest disturbances.

Iterative retraining can be viewed as a continuation of the single-date K–means classification with the addition of temporal and spatial (i.e., based on the spatial extent of spectral classes) information. Similar to when using the K–means classifier, the user can either set a fixed number of iterations or use a stopping criterion. In the latter case, the algorithm can be set to stop the retraining process if it no longer increases the number of TCS pixels. Considering that iterative retraining has the greatest impact on the processing time and that the third iteration typically brought very minor improvements, the algorithm was set to stop retraining after two iterations. Two iterations of retraining helped increase the number of detected TCS pixels by up to 70 %.

For the next step (primary change pixel detection) to work, no TCS pixels should be detected in areas disturbed during the processing time window period. Such areas indeed remained free of TCS pixels, even after multiple iterations of retraining. After two iterations, the resulting primary change pixels captured 40.8 % of the reference 1985 – 2015 cutblock pixels with a temporal agreement of ± 2 years, compared to 36.1 % without retraining. However, the impact on the final results (filtered secondary change pixels) was smaller, with a 2.1 percentage point increase in detected VRI pixels and a 0.024 percentage point increase in false detections over the same period (1985 – 2015).

As a side note, temporal inconsistencies in classification maps are a major source of error in traditional bi-temporal post-classification change detection techniques (Pontius and Lippitt, 2006; Burnicki et al., 2007). Although UC–Change is also a classification-based technique, it is much less sensitive in this regard, and not just because of the iterative retraining step described above. Firstly, it uses a large or very large number of image combinations (more than 30 sets of images in temporally dense datasets) to extract the initial set of change pixels (i.e., primary change pixels; §3.3) for a particular image acquisition date, instead of only two images. Secondly, UC–Change uses two additional steps to increase the number of change pixels, and then filter those detected change pixels, as described in §3.4 and §3.5.

3.3. Primary change pixel detection

The process of TCS pixel detection is repeated for every image in the time series. Once TCS pixels are detected for a new set of images, disturbances that occurred between the acquisition dates of the current and previous image are now designated as stable (TCS) areas. Primary change pixels are then extracted by subtracting old TCS pixels (i.e., temporally overlapping sets of TCS pixels detected for preceding images) from the new ones (Fig. 7d).

The main purpose of primary change pixel detection is to automatically generate training samples for the next step (secondary change pixel detection). Therefore, it is important that they represent only forest disturbances of interest. However, in addition to true disturbances, some pixels representing undisturbed forested areas can also be mistakenly labeled "change" pixels if the algorithm failed to detect them as stable in previous sets of images due to image noise and other factors. These mislabeled "change" pixels are usually spread out and can be removed by deleting change pixels that are not spatially clustered or occur in groups smaller than *40 Landsat pixels* (3.6 ha). While smaller patches of disturbed forest (< 3.6 ha), which account for 0.9 % (by area) of all clearcuts and 0 % of fire scars in the study area (according to the VRI and NBAC data), are not represented by the resulting set of primary change pixels, the removal of non-clustered pixels is a sacrifice worth making as it removes the vast majority of false detections. This value (40 pixels) provided clean and representative training areas for the next step, which is the secondary change pixel detection.

3.4. Secondary change pixel detection

After TCS and primary change pixels are detected for every image in the time series, more changes can be detected by performing a series of multitemporal classifications (MCs) using the same temporal subsets of images that were used in the TCS pixels detection step (Fig. 7e). For every such subset, UC–Change uses a stack of the corresponding four satellite images and training data consisting of TCS pixels representing areas that did not change during the corresponding time period and primary change pixels representing areas that did. The resulting MC map contains multiple stable and change classes, and each of the latter has an associated date of change. Consecutive temporal subsets of images overlap in time and, therefore, share some of the change classes. Combining same-date change classes from multiple consecutive MC maps produces secondary change pixels for each acquisition date. Because the same disturbance event may be assigned a different change date in different multitemporal classification maps, secondary change pixels are dated based on the majority rule. Interestingly, this approach (i.e., combining numerous MC maps based on the majority rule) can sometimes accurately predict the date of disturbance even for cutblocks and fire scars obscured by clouds or smoke in the year of change.

The Maximum Likelihood classifier was found well suited for multitemporal classification, as it was not prone to overfitting and was hundreds of times faster than the other two classifiers tested (Support Vector Machines and Neural Network). It was important that the data of resulting classes were normally distributed to prevent classes containing a mixture of disturbed and undisturbed forest. The Maximum Likelihood classifier aligned with this requirement due to its parametric statistical properties. In addition, because the same approach was used to map other parts of the province of British Columbia on a single computer, the processing speed was important.

As mentioned previously, while some intermediate steps in UC-Change involve supervised classification concepts, they do not rely on manually selected training samples but instead use automatically generated ones. Therefore, UC-Change is a fully unsupervised change detection method.

3.5. Object-based filtering of change pixels

Although the multitemporal classification step described in Section 3.4 substantially improves the detection rate, sometimes doubling the number of correctly detected reference change pixels, it can also introduce unwanted false detections of disturbance. Similar to the primary change pixel detection step, clusters smaller than 40 Landsat pixels (3.6 ha) were removed first. Doing so decreased the detection of reference cutblock pixels by 0.3 % while reducing false positives and non-target disturbances (i.e., not cutblocks or fire scars) by nearly 20 % compared to results produced with the minimum cluster size set to 20 pixels.

Once non-clustered change pixels are removed, change maps are segmented into individual spatial clusters and the distribution of spectral classes before and after disturbance are analyzed for every cluster. Spatial clusters that do not have substantially different pre- and post-disturbance distributions of spectral classes are rejected as described in the next paragraph. This filter requires ordinal categorical input with consistent class numbering. To meet this requirement, spectral classes in the single-date classification maps were ranked based on their average SWIR-2 pixel values. The SWIR-2 bands were chosen for three main reasons: 1) SWIR bands are the most important spectral bands in forest disturbance mapping (Zhu and Woodcock, 2014; Hermosilla et al., 2015a; Cohen et al., 2018); 2) SWIR-2 is least affected by thin clouds and haze compared to other optical spectral bands; and 3) SWIR-2 bands of the Landsat (TM, ETM+, and OLI) and Sentinel–2 sensors have relatively similar spectral response curves (Fig. 4). Such a ranking system assigns the highest rank to spectral classes containing disturbances and the lowest rank to classes representing dense forest. The ranking is relative and, therefore, does not require atmospheric correction to produce consistent results across the time series. The NIR-2 band was used for Landsat MSS data due to the lack of SWIR bands.

The UC–Change technique uses cumulative percentage of dominant pre- and post-disturbance spectral classes to remove spatial clusters of change pixels that do not represent clearcuts or fire scars. Dominant pre-disturbance classes are classes that, when combined, make up at least 90 % of clustered pixels in images acquired within four years before disturbance. In the study area, undisturbed coniferous stands were represented by Classes 1 to 5 in the 10-class SWIR-2-ranked classification maps. Dominant post-disturbance classes are determined the same way, but using images acquired within four years *after* disturbance. In the study area, most cutblocks and fire scars were represented by Class 10 and Classes 7 – 10, respectively (Fig. 8). A filter threshold of 2 was used in this study. This means that the percentage of dominant pre-disturbance classes must drop twofold and the percentage of post-disturbance classes must increase twofold following a disturbance event in order for the cluster of pixels to be labelled "disturbed". This threshold was determined to produce the lowest commission and omission errors in the study area and other test sites across Canada. A stricter threshold would not affect the detection of clearcuts, because the distribution of spectral classes changes dramatically. However, unlike clearcuts, fire scars are more heterogeneous and sometimes contain a similar set of spectral classes to the forest they replace. Therefore, depending on the area being mapped, the filter threshold may need to be adjusted to balance false detection and fire-scar detection. Since object-based filtering is the last step, and that it requires little time to complete, the user can quickly compare results across multiple threshold values. This process could potentially be automated, as most false detections occur near water bodies and in high-elevation areas – locations where forest harvest is either prohibited or is not feasible, and where forest fires are less common. The algorithm could incorporate a mechanism to minimize change detections in such areas by comparing to that in flatter areas away from water bodies, and adjusting the threshold accordingly.

3.6. Considerations for accuracy assessment

Due to the extensive study period spanning almost five decades, traditional accuracy assessment methods (e.g., error matrices) would introduce biases tied to sample distribution (i.e., imbalance between "change" and "no change" reference pixels) and temporal scope. For instance, longer study periods inherently include more change pixels, increasing the likelihood of false detections. This skews error metrics: omission errors (failure to detect true changes) may appear artificially low due to incidental detections of events like harvests or fires, while commission errors (false alarms) rise as spurious changes accumulate. While we provide an error matrix for a shorter common period (2002 – 2015) to enable comparison with other studies, our primary approach prioritizes temporal agreement between detected changes and reference data.

This time-aware method evaluates whether detected disturbances align with reference dates within defined tolerance windows (±1 or ± 3 years). By excluding detections outside these windows, we mitigate false positives and chance detections. The disparity between ± 1-year and ± 3-year agreement rates quantifies detection lag: the temporal offset between the first image in which a forest disturbance is first detected and the actual date of the disturbance.

Given the large size of the reference datasets (e.g., 8,111 VRI cutblock polygons representing 86.3 % of all clearcuts in the study area), any observed differences in results were attributed to the distinct approaches used by the change detection methods and the specific parameter settings (e.g., change thresholds) applied during map generation, rather than random chance or dataset limitations. Hence, it was deemed unnecessary to use Kappa statistics and confidence interval measures. Finally, VRI cutblock polygons were distributed evenly throughout the study area; therefore, no sampling design was employed (all reference cutblock data were utilized).

The time-aware evaluation approach used in this study produces easily interpretable results and offers a comprehensive understanding of the spatiotemporal accuracy of the maps; however, two considerations are necessary. Firstly, the distribution of forest fires was uneven in time and space. For this reason, the percentage of detected reference fire-scar pixels will be reported for individual years. Secondly, using protected areas to assess false detection of change was not ideal, as these areas are not fully representative of the study area, often being established around water bodies and in mountainous areas, where false detection of change is more likely due to the dynamic nature of wetlands and movements of the snow line, respectively. However, due to the lengthy time span and the fact that the same approach was applied to many other areas, it was considered the most practical and reproducible solution, despite the anticipated possible overestimation of false positives.

4. Results

4.1. Cutblock detection

Overall, UC–Change detected 85.2 % of all 1974 – 2018 VRI cutblock pixels with a temporal agreement of ± 1 year; 88.8 % were detected at ± 2 years and 90.3 % at ± 3 years (Table 1). In this particular study area, UC–Change outperformed all other techniques used for comparison across almost all years in the time series (Fig. 9). For the period 1985 – 2018, UC–Change mapped 86.0 % of forest harvest pixels using June 1 – September 20 data (78.4 % using July 1 – August 31 data). In comparison, C2C detected 58.8 % of reference cutblock pixels in the same period using July 1 – August 31 data. Combined UC–Change – C2C map contained 89.3 % of 1985 – 2018 VRI pixels at ± 1 year (93.6 % at ± 3 years). In terms of cutblock detection, the GFC and CCDC maps were more similar to the UC–Change map, except for the period 2012 – 2015, where the GFC map detected 26.7 % – 35.3 % fewer VRI pixels. Surprisingly, GFC and CCDC had the longest detection lag despite the fact that they used the most temporally dense datasets: they detected 9.1 % and 11.9 % more cutblock pixels at ± 3 years than at ± 1 year, respectively. For comparison, the difference for the same period was less than 5 % in the UC–Change and C2C maps.

4.2. Detection of fire scars

UC-Change achieved satisfactory results in the detection of fire scars (Table 2). It detected 70.9 % of reference 1987 – 2018 forest fire pixels at a ± 1-year agreement (72.3 % at ± 3 years), while C2C detected 54.6 %. UC–Change, CCDC, and GFC mapped 71.0 %, 37.4 %, and 67.2 % of pixels in the reference 2006 – 2018 forest fire polygons at a ± 1-year agreement and 72.4 %, 44.4 %, and 71.5 % at ± 3 years, respectively.

4.3. False detection of change

The C2C map had very few false cutblock pixels in protected areas (0.14 % of the total protected area) (Table 3). In contrast, UC–Change mapped 1.5 % of protected areas unaffected by forest fires as disturbed between 2002 and 2015 inclusive, indicating more errors of commission. As in the CCDC map, most false detections were along rivers and around lakes and ponds due to long-term fluctuations (4 – 5 years) in water level and wetland conditions. A total of 4.6 % and 8.9 % of protected areas were labelled disturbed in the CCDC and GFC maps, respectively.

Outside protected areas, UC–Change, C2C, CCDC, and GFC found change in 23.0 %, 21.6 %, 21.1 %, and 28.6 % of pixels in the 2002 – 2015 data, respectively. C2C mislabelled large portions of the 2017 wildfire as 2015 fires; without these mislabelled pixels, the percentage for the C2C map would reduce to just 14.9 %. According to the VRI and NBAC data, 14.9 % and 6.4 % of the unprotected area was harvested and burned during that period, respectively, resulting in a total of 20.2 % of the area affected by stand-replacing disturbances, including areas disturbed twice (fire and harvest). Considering that the VRI dataset contains approximately 90.6 % of cutblocks that occurred in the study area during the period 2002 – 2015 (based on our own set of reference cutblock polygons), the actual percentage of harvested and/or burned area would be expected to greater than 20.2 %. Therefore, the UC–-Change approach likely produced a better estimation of the total area affected by logging and forest fires compared to other methods considered. The CCDC estimate is not directly comparable in this context, as it may include other types of disturbances, such as MPB outbreaks.

Table 4 shows a confusion matrix for the two classes in the 2002 – 2015 change maps: "change" and "no change". Because the number of reference pixels for the "change" class (VRI + NBAC = 1,226,431) was

Table 1
Percent reference clearcut pixels detected at a temporal agreement of ± 1 and ± 3 years. The available CCDC and GFC maps covered limited time periods. UCC = UC–Change.

Years	Detection at ± 1 year (%)				Detection at ± 3 years (%)				Total area of VRI polygons (km^2)
	UCC	C2C	CCDC	GFC	UCC	C2C	CCDC	GFC	
1974–2018	85.2	–	–	–	90.3	–	–	–	2,752.0
1974–1984	76.3	–	–	–	85.1	–	–	–	245.4
1985–2018	86.0	58.8	–	–	90.8	63.6	–	–	2,506.6
2002–2018	87.1	54.5	74.0	70.4	91.5	59.1	82.8	76.8	1,383.4

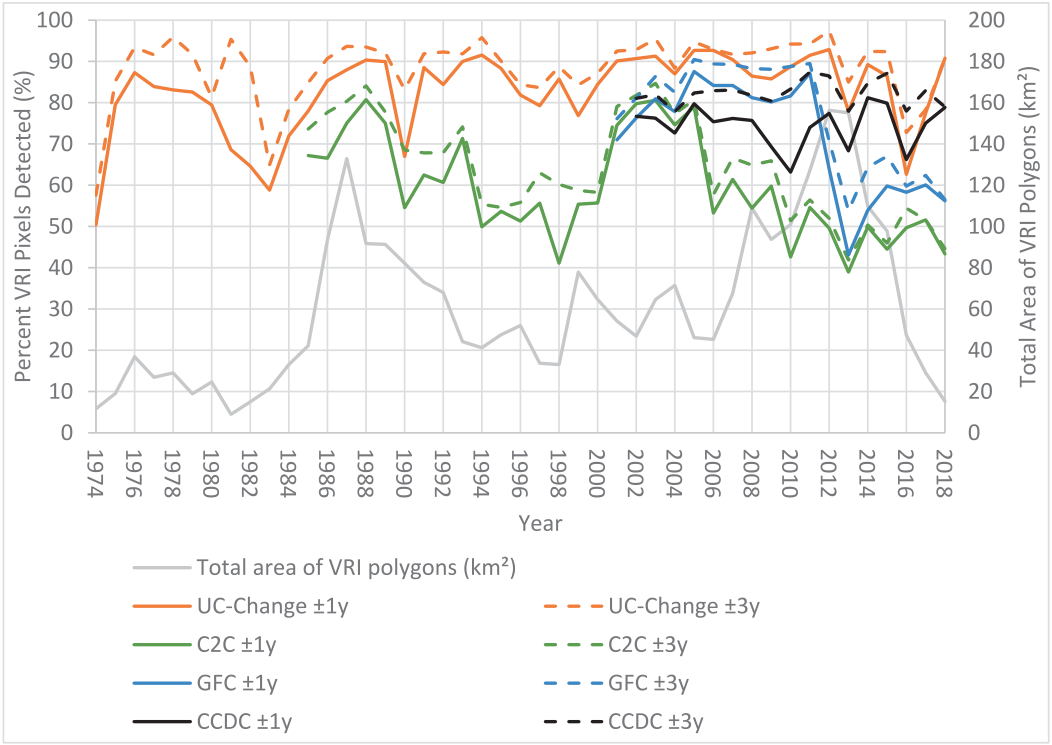


Fig. 9. Percent reference clearcut pixels detected at a temporal agreement of ± 1 year (solid lines) and ± 3 years (dashed lines). The grey line represents the total area of VRI clearcut polygons per year.

Table 2
Total area of the reference forest fire polygons per year and percent of the area detected by the four change detection techniques. The table only shows the years for which the total size of reference polygons was greater than 1 km^2. UCC = UC–Change.

Year	Detection at ± 1 year (%)				Detection at ± 3 year (%)				Total area of NBAC polygons (km^2)
	UCC	C2C	CCDC	GFC	UCC	C2C	CCDC	GFC	
1987	0.6	14.9	–	–	2.0	15.7	–	–	1.6
1988	92.7	87.9	–	–	94.9	88.7	–	–	1.4
1990	38.1	26.7	–	–	41.0	28.5	–	–	3.2
1995	56.3	58.4	–	–	56.4	58.4	–	–	19.2
2006	93.4	89.8	41.6	76.9	93.5	89.9	45.2	81.1	149.5
2009	68.2	60.0	39.4	54.1	69.4	66.1	43.9	57.3	53.9
2010	40.2	45.1	29.3	44.4	45.5	47.4	42.8	52.7	112.5
2014	87.5	76.2	37.4	50.6	87.6	79.3	41.7	57.3	134.3
2016	7.8	0.0	21.2	47.0	8.8	17.1	32.5	47.4	2.1
2017	72.1	51.3	38.4	69.1	73.5	69.9	45.8	73.2	3232.9
2018	44.0	72.6	22.9	57.4	44.4	72.9	24.7	60.0	193.9
All	70.9	54.6	37.4	67.2	72.3	70.3	44.4	71.5	3904.5

Table 3
False detection of cutblocks (commission errors) in the UC–Change and C2C maps for the period 2002 – 2015. Mislabelled 2017 forest fire pixels were removed from the C2C map. The CCDC, and GFC maps may contain disturbances other than logging and forest fires and were added for comparison only.

Percent area, excluding agriculture and water bodies, mapped as disturbed…	UC–Change	C2C	CCDC	GFC
…in protected areas unaffected by forest fires (out of 186.7 km^2)	1.45 %	0.14 %	4.62 %	8.91 %
…outside protected areas (out of 9684.8 km^2)	23.0 %	14.9 %	21.1 %	28.6 %

almost six times greater than for the “no change” class (protected areas unaffected by wildfires = 207,424), the imbalance between the sample distribution and the true distribution of classes would lead to biased overall accuracy estimates. Therefore, area weights were applied to reflect the true proportions of “change” (VRI + NBAC = 20.9 %) and “no change” (79.1 %) in the study area, resulting in overall accuracies of 96.2 %, 92.3 %, 91.3 %, and 90.3 % for UC–Change, C2C, CCDC, and GFC, respectively. Note that the CCDC map includes changes caused by factors other than harvest and wildfires; consequently, its results likely overestimate the false detection of change.

4.4. Visual analysis

A visual comparison of the maps showed that the C2C map contained much fewer clearcuts in the Very Dry Cold subzone of the Sub-Boreal Pine – Spruce biogeoclimatic zone than the UC–Change map (Fig. 10). In this subzone, C2C detected only 30.5 % of 1985 – 2015 VRI clearcut pixels compared to 63.1 % in the rest of the study area. GFC and CCDC also struggled to detect cutblocks in this subzone (59.0 % and 65.0 % of 2002 – 2015 VRI pixels detected in the subzone and 72.8 % and 74.2 %

Table 4
Area-weighted confusion matrix for the period 2002 – 2015. UCC = UC-Change.

Map		Reference data (% of the Study Area)			
		Harvest and Fire	No Harvest, No Fire	Total % (Map)	User's Accuracy (%)
Change	UCC	18.2	1.1	19.4	94.1
	C2C	13.3	0.1	13.4	99.2
	CCDC	15.9	3.7	19.5	81.3
	GFC	18.2	7.0	25.3	72.1
No change	UCC	2.7	78.0	80.6	96.7
	C2C	7.6	79.0	86.6	91.2
	CCDC	5.0	75.5	80.5	93.8
	GFC	2.7	72.1	74.7	96.4
Total % (Reference)		20.9	79.1	100.0	Overall (%)
Producer's Accuracy (%)	UCC	87.2	98.5		96.2
	C2C	63.6	99.9		92.3
	CCDC	76.0	95.4		91.3
	GFC	87.1	91.1		90.3

outside, respectively). In contrast, UC–Change had nearly the same detection rate throughout the study area (84.7 % and 87.0 % at ± 1 year, respectively).

Clearcuts in the UC–Change map had a more homogenous, spatially contiguous appearance with almost no "salt-and-pepper" (speckle) effect compared to other maps (Fig. 11). Unlike the GFC map, UC–Change and CCDC accurately represented forest reserves inside clearcuts. In contrast, many cutblocks were absent or only partially present in the C2C map (e.g., Fig. 11).

5. Discussion

5.1. Performance of the UC–Change technique

The performance of the UC–Change technique was relatively consistent across time periods. Somewhat lower detection of clearcuts in the 1974 – 1983 time period, for which only Landsat MSS data were available, can be attributed to several factors: lack of SWIR bands, image quality (e.g., low signal-to-noise ratio and large geometric errors), and low spatial resolution (68 m × 83 m resampled to 60 m). Only 76.3 % of 1974 – 1984 cutblocks were detected and dated within ± 1 year from the corresponding VRI timestamps, while 85.1 % were detected at a ± 3-year accuracy, which indicates a delayed detection of clearcuts. These temporal inaccuracies in the UC–Change map were likely caused by the aggressive cloud masking that was applied to the input MSS data, which created large temporal data gaps between consequent observations.

The higher quality of Landsat 5 TM data and available cloud masks improved the temporal accuracy and overall detection of 1984 – 1998 disturbances (85 % of reference cutblock pixels detected at ± 1 year and 90 % at ± 3 years). The addition of Landsat 7 ETM + observations resulted in the highest agreement with the reference data, maintaining a detection rate close to 90 % (93 % at ± 3 years) between the years 2000 and 2012.

The results after 2013 had lower accuracies, despite the availability of high-quality Landsat 8 OLI and Sentinel–2 data. This was likely due to the inability of the algorithm to detect multiple disturbances if they occur within the processing time window, which was four years in this study. In this case, a 2017 wildfire engulfed more than a third of the study area. Because of the 2017 fire, fewer TCS pixels were detected in the 2013 – 2017 data, resulting in fewer primary change pixels and, consequently, secondary change pixels. Removal of post-2016 images from the dataset improved the detection of 2013 – 2015 clearcuts by 21.6 %, confirming the hypothesis that the relatively low performance in

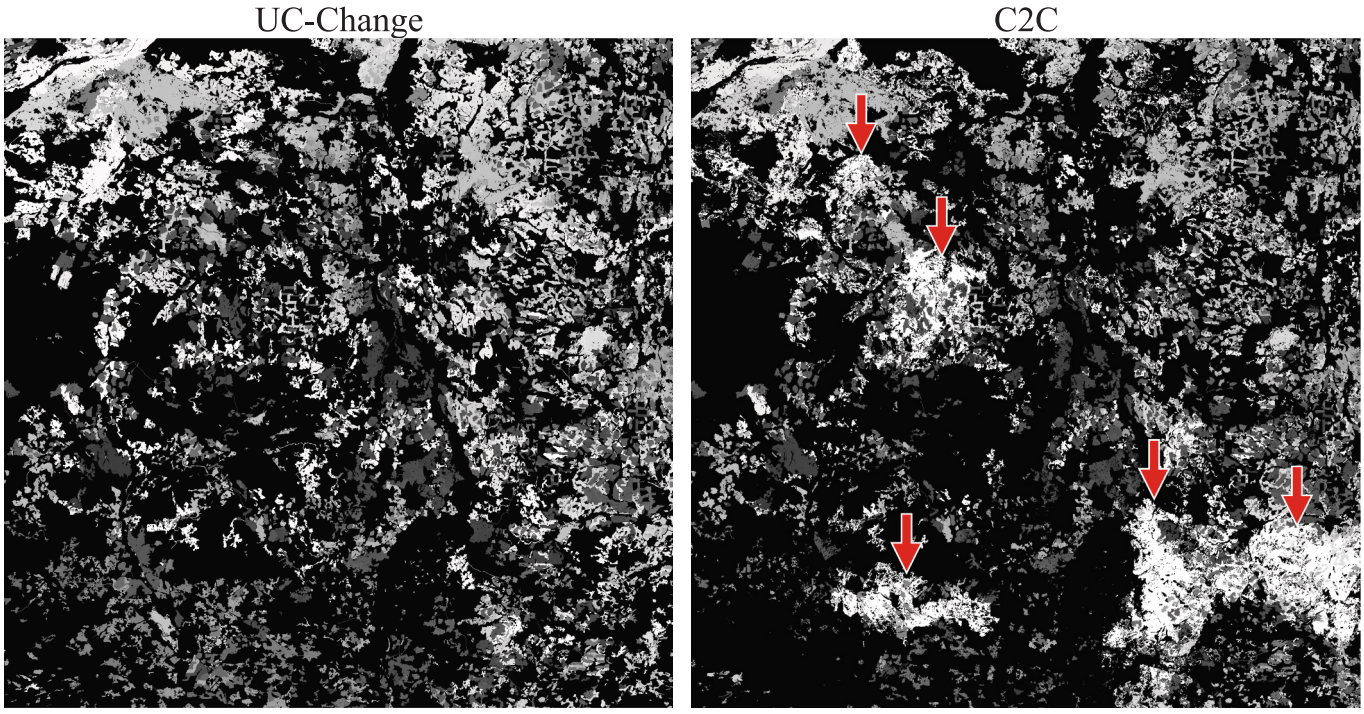


Fig. 10. UC–Change and C2C forest change maps of the entire study area (100 km × 100 km; north is up) for the period 1985 – 2015. This period is shown instead of 1985 – 2020 because the 2017 wildfire, which affected 32.3 % of the study area, would obscure cutblocks that occurred in the affected area before or after the fire, making them difficult or impossible to see. Portions of that wildfire were mislabelled by the C2C algorithm as 2015 fires (white patches indicated by red arrows). Brighter shades of grey indicate more recent changes. Note the differences in the number of detected cutblocks, especially in the Sub-Boreal Pine – Spruce bio-geoclimatic zone covering the south-west corner of the map. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

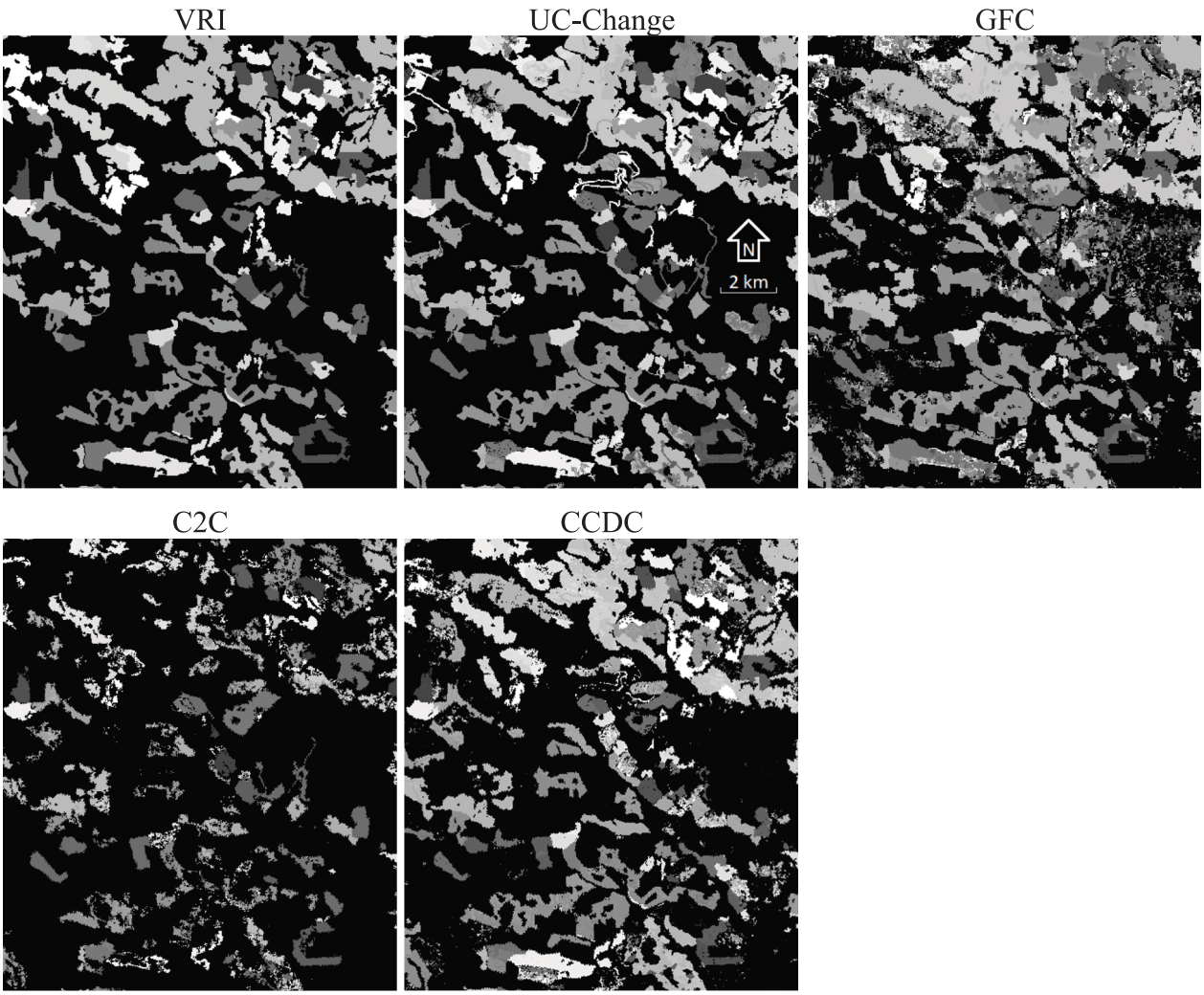


Fig. 11. 2001 – 2015 clearcuts and clearcuts with reserves (patches of undisturbed forest) as they appear in the reference data (VRI) and the five change detection maps compared in the paper. The UC–Change map was the most similar to the reference map. It accurately represented clearcuts with reserves and had a cleaner look overall, whereas other maps had a strong "salt-and-pepper" effect with many missing or only partially represented cutblocks. Brighter shades of grey indicate more recent changes.

this period was caused by the 2017 fire.

A second contributing factor to the relatively low detection of 2016 and 2017 cutblocks was the image-based nature (as opposed to pixel-based) of UC–Change. Large-area disturbances (e.g., a major forest fire), unmasked clouds and wildfire smoke, and agricultural land can affect the detection of disturbances in the surrounding areas by altering the distribution of spectral classes in the rest of the image. While the iterative retraining step (§3.2.2) mitigated this problem to some extent by making classification consistent among images, the image–based approach still led to some clearcuts being missed – even in areas unaffected by the 2017 fire.

UC–Change detected relatively fewer reference forest fire pixels (71.0 % at ± 1 year for the period 1987 – 2018) than reference cutblock pixels (86.0 %), and this was expected. Compared to clearcuts, fire scars are more heterogeneous and may contain unaffected trees, information that is not contained in the reference polygons. If the NBAC polygons represented only areas with fully burned canopies, the detection rate would likely be only slightly lower than that of cutblocks.

5.2. Comparison with existing change detection maps

The resulting UC–Change maps were different from the maps produced with the other change detection methods tested. The most obvious difference was the total area mapped as disturbed. UC–Change, C2C, CCDC, and GFC mapped 2228 km^2, 1445 km^2, 2050 km^2, and 2789 km^2 as disturbed during the period of 2002 – 2015, respectively. Such a large difference can be partially explained by the extensive MPB outbreak in the 2000 s, which killed most lodgepole pine trees in the area (BC Ministry of Forests, 2023a). The CCDC map does show some of the MPB damage, although the omission errors for this type of disturbance are very high (Ye et al., 2021), whereas the UC–Change map and the available C2C map do not. Based on visual interpretation, GFC also detected some MPB-affected areas.

The MPB outbreak can explain why the C2C technique detected so few 2006 – 2015 reference cutblocks pixels (38.8 % at a temporal agreement of ± 1 year). Salvage logging accounted for a significant proportion of wood harvested in the area during this period (Kurz et al., 2008). Because many trees were already dead, the difference between pre- and post-harvest NBR values in MPB-affected stands was too low for C2C to detect cutblocks. However, it is noteworthy that compared to UC–Change, this technique also detected fewer reference cutblock pixels before the outbreak: C2C mapped 67.8 % of 1985 – 1999 VRI cutblock pixels at a ± 1-year agreement, whereas UC–Change mapped 83.6 %. This indicates that UC–Change was more capable at detecting cutblocks in the study area, likely due to its ability to use more spectral bands and approximately double as many images per year (June – September

imagery as opposed to July and August). CCDC and GFC also used multiple spectral bands instead of a single spectral index (NBR) to detect change, leading to superior cutblock detection compared to C2C. If C2C used a more relaxed change threshold and then filtered out objects smaller than 3.6 ha, similar to UC–Change, it would have likely detected more cutblock pixels.

UC–Change also performed very well in the Very Dry Cold subzone of the Sub-Boreal Pine – Spruce biogeoclimatic zone, where the other four techniques (especially C2C) struggled the most. Due to the low soil moisture and stand density, undisturbed forest stands in this part of the study area have relatively low NBR values, making clearcuts and forest fires harder to detect for techniques that rely exclusively on this spectral index, such as C2C.

One of the main advantages of the UC–Change technique is its flexibility. It can use datasets containing any number of images per year and any number of spectral bands per image. Because the technique can use data acquired during any part of the growing season, the availability of cloud-free data was much less of an issue. June to September data provided a full coverage of the study area for most years in the time series. In contrast, the C2C map was produced using only data acquired during the peak of the growing season (July and August). This limitation likely affected the performance of the C2C algorithm in a negative way. For example, Hermosilla et al. (2015b) found that cloud-free July and August Landsat data covered only 85 % of the forested area of Saskatchewan on average per year during the period 2000 to 2010. It would be even more difficult to find sufficient amount of data for the years 1984 to 1998, for which only Landsat 5 TM imagery was used (as opposed to Landsat 5 and 7 for the period 2000 – 2010). However, UC–Change still detected substantially more cutblock pixels for this period than C2C when only July and August data were used (79.7 % vs. 54.9 %) instead of June 1st – September 20th data (84.5 %), which indicates that the narrow date range used by C2C was a significant but not the most important factor that affected their ability to detect forest harvesting in the study area.

The CCDC map had a much larger detection lag for some forest disturbances compared to the UC–Change and C2C maps. Unlike temporal trajectory analysis techniques based on linear regression (e.g., LandTrendr and C2C), CCDC does not produce a continuous trajectory where the segments representing pre- and post-disturbance states are connected. Instead, these segments are separated by breaks/gaps (Fig. 1). Longer periods of missing data result in longer breaks. In addition, CCDC requires six consecutive clear observations to confirm a change. Due to the limited availability of snow-free and cloud-free data in the study area, these six observations may span several years, resulting in a detection lag.

UC–Change, C2C, and GFC performed substantially better at detecting burned forests than CCDC. The CCDC method detected change in only 37.4 % of the reference forest fire pixels at a temporal accuracy of ± 1 year, compared to performance levels of 70.9 %, 54.6 %, and 67.2 % for UC-Change, C2C, and GFC, respectively. The CCDC detection rate increased to 44.4 % at ± 3 years and 47.8 % when the date of change was ignored. It is possible that the change magnitude in many of the spectral bands used by CCDC did not exceed the thresholds required to classify spectral changes as disturbance. The CCDC technique is univariate in nature and processes spectral bands separately before combining the results. In contrast, UC–Change uses all bands together and can detect disturbances that have a subtle effect on individual bands, such as fires in MPB-affected forests.

Using protected areas, such as provincial parks, to quantify commission errors helped identify a problem in the current version of the UC–Change algorithm and available CCDC maps. Most false detections in the UC–Change and CCDC maps were found in wetlands, where spectral changes resulting from interannual variation in water levels and precipitation were often misidentified as forest disturbances. Coincidentally, most parks in the study area were established around large lakes and contain many wetlands. As a result, there were relatively more errors of commission in parks than in the rest of the study area. Based on a visual analysis, false detections in the C2C and GFC maps occurred mostly in forested areas and were more evenly spread out throughout the study area.

6. Conclusions

The UC–Change technique represents a fundamentally new approach to forest disturbance mapping. UC–Change uses classification images to detect and date stand-replacing disturbances, which makes it primarily independent of spectral resolution and data type, be it radiance or reflectance. For the same reason, the algorithm can use any number of spectral bands of any sensor to detect disturbances. This flexibility contrasts with many existing change detection methods that use only one band or spectral index or the same set of bands for all sensors, falsely assuming identical spectral characteristics across different sensors.

The technique is also very flexible in terms of requirements to the timing of input data. While other methods either use only mid-summer data (e.g., LandTrendr and C2C use July and August data; Kennedy et al., 2010; Hermosilla et al., 2015a) or require dense time series (e.g., seven clear observations per year; Zhang et al., 2021), the proposed method can use any number of non-winter images. Therefore, data availability is much less of an issue for UC–Change compared to these two approaches. Furthermore, unlike existing classification-based methods, UC–Change uses an unsupervised approach that requires little to no user input (parameter adjustment may be required when applying the algorithm to new geographical areas). These two advantages potentially make UC–-Change more suitable for mapping forest disturbance in areas and time periods with limited satellite and reference data availability.

Finally, UC–Change does not require atmospheric correction or radiometric normalization, facilitating the use of Landsat MSS data and extending the temporal range of forest disturbance mapping back to 1972. The UC–Change technique detected the vast majority of 1974 – 1984 clearcuts in Landsat MSS images (85.1 % at ± 3 years), which demonstrates the value of MSS data for understanding forest cover dynamics. However, the quality of the data was inconsistent and required some manual image selection. Therefore, it is recommended that data providers improve image quality, as well as cloud and cloud shadow detection and geometric correction algorithms, for Landsat MSS imagery.

The study used Landsat and Sentinel 2 data over a 100 km × 100 km area south of Prince George, British Columbia, featuring diverse forests from sparse pine to dense spruce. UC–Change detected 85.2 % of reference clearcut pixels (1974–2018) and 71.0 % of fire scar pixels (1987–2018) at a ± 1-year agreement (90.3 % and 72.4 % at ± 3 years). Compared with C2C, CCDC, and GFC maps, UC–Change showed more consistent performance over space and time, detecting more cutblock and fire-scar pixels, especially in open coniferous forests and areas affected by MPB, albeit with more false detections compared to C2C.

Interestingly, the performance of the technique was unaffected by a major MPB outbreak. This illustrates the robustness of the UC–Change approach in terms of detecting disturbances of interest — namely, cutblocks and fire scars — including salvage harvesting and fires in stands killed by the MPB, while effectively ignoring MPB damage itself. If types of disturbance such as MPB outbreaks were of interest to be detected via UC-Change, the unsupervised classifier and subsequent steps could be adapted to separate those disturbances, including with respect to their unique temporal signatures.

The two main limitations of UC–Change stem from its image–based classification and the length of its processing time window. Scene-wide classification can be affected by large disturbances or unmasked atmospheric artifacts that alter spectral class distributions across an image, although the iterative retraining step helps mitigate this. Additionally, disturbances that occur close together in time may go undetected if they fall within the same multi-year window. Increasing data availability from current and upcoming sensors will support shorter time windows

and enhance the technique's capacity to capture closely spaced disturbance events.

The next research steps for UC–Change include testing the technique over other regions in Canada and elsewhere, with further algorithm refinements, such as change attribution, automatic parameter adjustment, automatic geometric correction and filtering of Landsat MSS data, and support for data of various spatial resolution without resampling (e.g., 10-m and 20-m Sentinel–2 data and 15-m Landsat panchromatic data). False detection of change in wetlands is also an issue that needs to be refined. Finally, the flexibility and robustness of UC–Change may create synergies with other powerful approaches, such as using canopy reflectance models in forest structural classification (Soenen et al., 2005, 2008, 2010; Peddle et al., 2004, 2011).

CRediT authorship contribution statement

Ilia Parshakov: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Derek Peddle:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition. **Karl Staenz:** Writing – review & editing, Resources, Funding acquisition. **Jinkai Zhang:** Writing – review & editing. **Craig Coburn:** Writing – review & editing, Funding acquisition. **Howard Cheng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was conducted within the Alberta Terrestrial Imaging Centre (ATIC) [now Institute for Geospatial Inquiry, Instruction & Innovation (i4Geo)] and the Department of Geography and Environment at the University of Lethbridge and supported primarily by a Natural Sciences and Engineering Research Council of Canada Collaborative Research and Training Experience grant entitled Advanced Methods, Education and Training in Hyperspectral Science and Technology (NSERC CREATE AMETHYST; P.I.'s Dr. P. Teillet, Dr. K. Staenz and Dr. D. Peddle). We acknowledge additional support provided by the University of Lethbridge. Dr. H. Peter White and Dr. Ron Hall of Natural Resources Canada and Dr. Dongmei Chen of Queen's University are thanked for reviewing an earlier version of this work. We thank the journal's Editor and anonymous reviewers for their comments and suggestions that helped us improve the content and clarity of this work.

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